

Personalized 30-day cybersecurity engineer roadmap for a student with C++, SQL, HTML/CSS skills, 2 hours daily, video learning style

Cybersecurity engineer — 30-day roadmap

Student · Beginner · 2 hrs/day · Video-first

C++ SQL HTML/CSS

30
days

60+
hrs learning

4
projects

1
cert track

Week 1 — Foundations Week 2 — Networking Week 3 — Offense Week 4 — Defense & career

Security foundations & your existing skills
Days 1–7 · CIA triad, Linux, cryptography — leveraging your C++ background

Your C++ helps here — buffer overflows and memory exploits will make sense faster than for most beginners.

DAILY TASKS

- Day 1

CIA triad, threat actors & attack types (video)
- Day 2

Kali Linux setup + terminal essentials
- Day 3

How C++ relates to memory exploits (buffer overflow intro)
- Day 4

Cryptography basics — hashing, AES, TLS (video)
- Day 5

Password security, auth methods, MFA
- Day 6

OverTheWire: Bandit levels 0–10
- Day 7

Week review + quiz yourself

VIDEO-FIRST RESOURCES

- Video

Professor Messer Security+ (YouTube, free)
- Platform

TryHackMe — Pre-Security path
- Practice

OverTheWire: Bandit wargame
- Video

NetworkChuck — Kali Linux setup (YouTube)

WEEK PROJECT

Kali lab + C++ exploit demo

Set up VirtualBox with Kali Linux. Write a simple C++ program that intentionally has a buffer overflow, then explain in a README why it's dangerous and how ASLR mitigates it.

Your outcome after 30 days

Understand attacks, defenses, and how your C++ knowledge maps to security exploits

Hands-on practice with Nmap, Wireshark, Burp Suite, and Metasploit

Completed 4 portfolio projects including a vulnerability scanner and CTF writeups

SQL injection expertise — your SQL background gives you a real edge here

Ready to sit CompTIA Security+ or pursue an internship in a SOC role